

## WEEKLY REPORT 4

### GROUP MEMBER AND ROLL NO

- Yash Kirti AU2340149
- Yashvi Modi AU2340116
- Priyanshu Rupala AU2340207
- Tathya Parikh AU2340151

### **Week: 2 (Super-Resolution Implementation Phase) Project Title: Super Resolution using Classical Interpolation Methods**

#### Objective

This week was aimed at adding a state-of-the-art classical SR algorithms to move beyond interpolation-based upscaling to high-fidelity image reconstruction. The overall aim was to bring together edge-directed, iterative, and frequency-domain refinement model into an integrated pipeline that can be successfully used to reconstruct finer details using satellite LR images.

#### **Work Done**

##### a. Application of Edge-Directed Interpolation (NEDI).

Our version of NEDI algorithm (New Edge-Directed Interpolation) was simplified. It compares the local edge structures and maintains the directionality of the edges throughout the interpolations.

To increase edge regions, Canny edge detector and Gaussian weighting were employed to improve the computational efficiency.

This made sure that roads, coastline and building edges were sharper in the reconstructed images of SR.

It was visually verified that there was a significant enhancement of edge clarity over Bilinear and Bicubic techniques.

##### b. Iterative Back projection (IBP) Refinements

Once I had the initial SR image produced with the help of NEDI, IBP was used to refine the image.

This is done by downscaling the SR image to LR, comparing with original LR, and computing the error and the process of back-projecting the error is repeated.

The reconstruction artifacts are eliminated by each iteration and the data fidelity is enhanced.

Using experimentation, we had 4 iterations as optimum on images of 1024x1024 to 2048x2048.

IBP helped cut down on aliasing to a great degree and offer sharper texture on the terrain.

### c. Wavelet-Based Detail Enhancement.

The last process of the SR pipeline involved Haar wavelet decomposition. The picture was broken down into 4 sub-bands (LL, LH, HL, HH).

Factors of high-frequency components (LH, HL, HH) were increased 1.31 to 1.5 depending on image size.

Enhanced sub-bands were again recombined to create the final image which is super-resolved.

This improved textures like patterns of forest and cracks in soil and ripples of water giving a rich output in terms of visuals.

## Observations

| Method   | Avg PSNR (dB) | Avg SSIM | Observations                                     |
|----------|---------------|----------|--------------------------------------------------|
| Nearest  | 28.4          | 0.44     | Blocky artifacts, poor fidelity                  |
| Bilinear | 29.6          | 0.50     | Smooth transitions, blurred textures             |
| Bicubic  | 31.0          | 0.56     | Balanced results but loss of high-frequency data |

|                         |             |             |                                                        |
|-------------------------|-------------|-------------|--------------------------------------------------------|
| NEDI + IBP +<br>Wavelet | <b>33.9</b> | <b>0.69</b> | Clear textures, sharp edges, visually<br>closest to HR |
|-------------------------|-------------|-------------|--------------------------------------------------------|

### Key findings:

Combined approach produced the best PSNR and SSIM of all the datasets.

This hybrid approach was the most beneficial to urban and agricultural areas.

There were some ringing artifacts observed along strong contrast boundaries - this may be possibly improved in the post-processing.

### Challenges

Multiple repeated IBP took more computational time.

Wavelet enhancement sometimes enhanced small noise.

### Next Week Plan

Accomplish the comprehensive comparative assessment of all 10 pictures.

Make bar graphs and visual panels of HR, LR, Bicubic and Final SR outputs.

Start writing Results and Discussion of the report.